

Heat Transfer Homework

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Problem Set Number: #3

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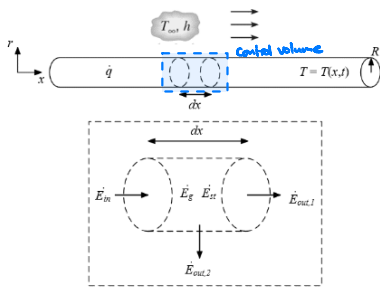
Known: environment $\rightarrow T_{\infty}$ dimensions $\rightarrow$ radius R
length L
x-axis through center

volumetric heat generation $\rightarrow \dot{q}$ [W/m^3]

for c) $T(0,t) = T_i$
 $T(L,t) = T_{\infty}$, h
 $T(x,0) = T_i$

Find: a) energy balance eq., express each term in terms of T
b) derive the governing eqs of the rod & simplify
c) BC & IC for the governing eq.

schematic:



assumptions: very small radius, k, ρ, c_p are constant,
no radiation effects

analysis:

$\rightarrow \dot{q}(x) + \dot{q}(dx) = \dot{q}(x) + \frac{d\dot{q}}{dx} dx$

A $\dot{E}_{in} + \dot{E}_g - (\dot{E}_{out,1} + \dot{E}_{out,2}) = \dot{E}_{st}$

↑ conduction ↑ convection

$-kA dx \frac{\partial T}{\partial x} + \dot{q} A dx - \left(-kA dx \frac{\partial T}{\partial x} - kA dx \frac{\partial^2 T}{\partial x^2} + h(T - T_{\infty}) A_s \right) = \rho c_p A dx \frac{\partial T}{\partial t}$

↑ surface area

$$kA dx \frac{\partial^2 T}{\partial x^2} + \dot{q} A dx - h(2\pi R dx)(T - T_{\infty}) = \rho c_p A dx \frac{\partial T}{\partial t}$$

B $\frac{kA dx \frac{\partial^2 T}{\partial x^2}}{A dx} + \frac{\dot{q} A dx}{A dx} - \frac{h(2\pi R dx)(T - T_{\infty})}{\pi R^2 dx} = \frac{\rho c_p A dx \frac{\partial T}{\partial t}}{A dx}$ divide by area

$$k \frac{\partial^2 T}{\partial x^2} + \dot{q} - \frac{2h}{R} (T - T_{\infty}) = \rho c_p \frac{\partial T}{\partial t}$$

c BC $\Rightarrow T(0,t) = T_i, -k \frac{\partial T}{\partial x} \Big|_L = h(T(L,t) - T_{\infty})$

IC $\Rightarrow T(x,0) = T_i$

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Known: $L_A = 3\text{ m}$, $L_B = .1\text{ m}$, $T_\infty = 20^\circ\text{C}$, $T_w = 115^\circ\text{C}$
 $k_B = 30\text{ W/mK}$, $h = 500\text{ W/m}^2\text{K}$, $T_i = 100^\circ\text{C}$

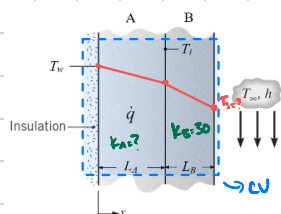
find: T_s , right side of wall B

$\dot{q}$ @ steady state

k_A of wall A

plot T profiles across walls A & B + important feature

Schematic:



assumptions: left surface of wall A is thermally insulated
 no $\dot{E}_g$ in wall B, no thermal resistance between walls, steady state

analysis:

$$A. \dot{q}_{\text{cond}} = \dot{q}_{\text{conv}} \Rightarrow \frac{-k_B(T_s - T_i)}{L_B} = h(T_s - T_\infty)$$

$$-T_s \left(\frac{k_B}{L_B} + h \right) = -T_\infty h - T_i \frac{k_B}{L_B} \quad \text{divide by } -1$$

$$T_s = \frac{T_\infty h - T_i \frac{k_B}{L_B}}{\frac{k_B}{L_B} + h} = \frac{(20)(500) - (100)\left(\frac{30}{.1}\right)}{\frac{30}{.1} + 500} = \boxed{50^\circ\text{C}}$$

$$B. \dot{q}_{\text{cond}} = \frac{-k_B(T_s - T_i)}{L_B} = -\frac{30}{.1}(50 - 100) = 15000 \text{ or } 15 \frac{\text{KW}}{\text{m}^2}$$

$$\dot{q} = \frac{\dot{q}_{\text{cond}}}{L_A} = \frac{15 \text{ KW/m}^2}{.3} = \boxed{50 \text{ KW/m}^3}$$

$$C. \int \frac{\partial^2 T}{\partial x^2} + \frac{\dot{q}}{k_A} = 0$$

$$\int \frac{\partial T}{\partial x} + \frac{\dot{q}x}{k_A} + C_1 = 0$$

$$T(x) + \frac{\dot{q}x^2}{2k_A} + C_1 + C_2 = 0 \Rightarrow T(x) = -\frac{\dot{q}x^2}{2k_A} + C_2$$

$\rightarrow T(0) = 0 \Rightarrow C_2 = T_\infty$

$$T(x) = -\frac{\dot{q}x^2}{2k_A} + T_\infty$$

BC $\Rightarrow \frac{dT}{dx} \Big|_{x=0} = 0$ (insulation temp), $T(0) = T_\infty$

$$T(x) = \frac{-\dot{q}x^2}{2k_A} + T_w \Rightarrow T_i = \frac{-\dot{q}L_A^2}{2k_A} + T_w$$

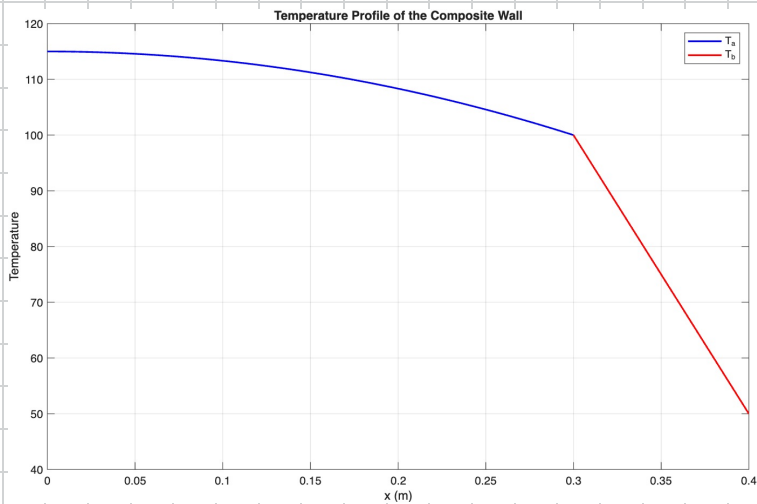
$$k_A = \frac{-\dot{q}L_A^2}{2(T_i - T_w)} = \frac{-(50 \times 10^3)(0.3)^2}{2(100 - 115)} = 150 \frac{\text{W}}{\text{mK}}$$

$$D \quad T(t) = T_A(t) + T_B(t) = \frac{-\dot{q}x^2}{2k_A} + T_w + \frac{q''(x-L_B)}{k_B} + T_i$$

$$q'' = \frac{k_B}{x-L_B} (T_B(t) - T_i)$$

$$T_B(t) = \frac{q''(x-L_B)}{k_B} + T_i$$

piecewise $x(t) \begin{cases} T_A: 0 \leq x \leq L_A \\ T_B: L_A \leq x \leq L_B \end{cases}$



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1 %% plot temperature profile
2 Tw = 115; Ti = 100;
3 ka = 150; kb = 30;
4 q_dot = 50e3; q_flux = 15e3;
5 La = 0.3; Lb = 0.1;
6
7 %domain and piecewise
8 x = unique([linspace(0, La, 300), linspace(La, La+Lb, 200)]);
9 xa = x(x>=0 & x<=La);
10 xb = x(x>=La & x<=La+Lb);
11
12 %equation
13 Ta = (-q_dot.*xa.^2) ./ (2*ka) + Tw;
14 Tb = (q_flux.*(xb - La) ./ -kb) + Ti;
15
16 %plotting
17 plot(xa,Ta,'b',xb,Tb,'r','LineWidth',1.5)
18 xlabel('x (m)')
19 ylabel('Temperature')
20 title('Temperature Profile of the Composite Wall')
21 legend('T_a','T_b')
22 grid on

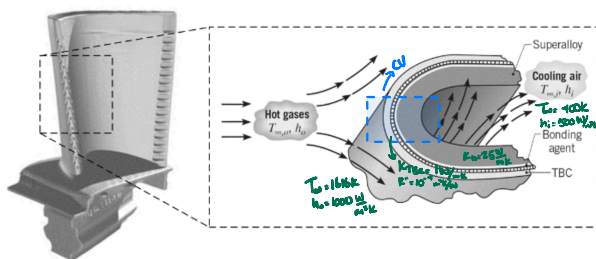
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Known: $k_b = 25 \frac{\text{W}}{\text{mK}}$, $k_{TBC} = 1 \frac{\text{W}}{\text{mK}}$, $T_{\infty,o} = 1616 \text{K}$, $T_{\infty,i} = 900 \text{K}$, $L_b = 5 \times 10^{-3} \text{m}$ for b) $L_{TBC} = 5 \times 10^{-3} \text{m}$
 $h_o = 1000 \frac{\text{W}}{\text{m}^2\text{K}}$, $h_i = 500 \frac{\text{W}}{\text{m}^2\text{K}}$, $T_{max} = 1200 \text{K}$
 $R'' = 10^{-1} \frac{\text{m}^2\text{K}}{\text{W}}$

Find: thermal resistance network, ext. surface temp. maintained below T_{max} for a) no TBC b) with TBC

Schematic:



assumptions: steady state, bonding agent thickness negligible, radiation negligible, black as plane wall

analysis:

A network $\Rightarrow$ $T_{\infty,o} \xrightarrow{R''_{conv,o}} T_o \xrightarrow{R''_{cond}} T_i \xrightarrow{R''_{conv,i}} T_{\infty,i}$

$$R_{total} = R''_{conv,i} + R''_{cond} + R''_{conv,o}$$

$$= \frac{1}{h_i \cdot A} + \frac{L_b}{k_b \cdot A} + \frac{1}{h_o \cdot A} \quad \text{multiply by } A$$

$$= \frac{1}{h_i} + \frac{L_b}{k_b} + \frac{1}{h_o}$$

$$R_{total} = \frac{\Delta T}{\dot{q}} \Rightarrow \dot{q} = \frac{T_o - T_i}{R_{total}} = \frac{1616 - 900}{\frac{1}{500} + \frac{5 \times 10^{-3}}{25} + \frac{1}{1000}} = 380 \frac{\text{KW}}{\text{m}^2}$$

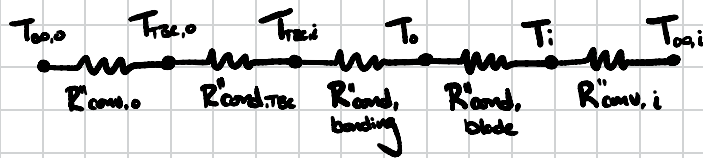
$\Downarrow$

$$\dot{q} = h_o(T_{\infty,o} - T_o) \Rightarrow T_o = \frac{\dot{q}}{h_o} + T_{\infty,o} = \frac{380}{1} + 1616 = \boxed{1236 \text{K}}$$

$$1236 \text{K} > T_{max} = 1200 \text{K}$$

cannot be maintained below T_{max}

B



$$R_{total} = R_{conv,o}'' + R_{cond,tec}'' + R_{cond,band}'' + R_{cond,blade}'' + R_{conv,i}''$$

$$= \frac{1}{h_o} + \frac{L_{tec}}{k_{tec}} + R'' + \frac{L_b}{k_b} + \frac{1}{h_i}$$

divide by A

$$q'' = \frac{T_{\infty,o} - T_{\infty,i}}{R_{tot}} = \frac{1616 - 400}{\frac{1}{1000} + \frac{5 \times 10^{-9}}{1} + 10^{-9} + \frac{5 \times 10^{-3}}{25} + \frac{1}{500}} = 320 \frac{KW}{m^2}$$

$$q'' = \frac{T_o - T_{\infty,i}}{R_{eq}''}$$

$$\Rightarrow T_o = q'' \cdot R_{eq}'' + T_{\infty,i}$$

$$= (320 \times 10^3) \left(\frac{5 \times 10^{-3}}{25} + \frac{1}{500} \right) + 400$$

$$= 1101K$$

$$1101K < 1200K = T_{max}$$

blade can be maintained below T_{max}